The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 February 2000.

INCH-POUND

MIL-PRF-19500/448B
3 February 1999
SUPERSEDING
MIL-S-19500/448A(USAF)
29 March 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPE 2N4405
JAN, JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, silicon, low-power transistors. Two levels of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Types	P_T 1/ T_C = +25°C	P_T 2/ T_A = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_{OP}
	W	W	V dc	V dc	V dc	A dc	°C
2N4405	5.0	1.0	80	80	5	0.5	-55 to +200

1/ Derate linearly, 28.6 mW/°C for $T_C \geq +25°C$.
2/ Derate linearly, 5.72 mW/°C for $T_A \geq +25°C$.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	h_{FE1} 1/	h_{FE3} 1/	$V_{BE(SAT)2}$ 1/	$V_{CE(SAT)3}$ 1/	C_{obo}	$\|h_{fe}\|$
	V_{CE} = 5 V dc I_C = 100 µA dc	V_{CE} = 5 V dc I_C = 150 mA dc	I_C = 500 mA dc I_B = 50 mA dc	I_C = 500 mA dc I_B = 50 mA dc	V_{CB} = 10 V dc I_E = 0 mA dc 100 kHz ≤ f ≤1 MHz	V_{CE} = 20 V dc I_C = 50 mA dc f = 100 MHz
			V dc	V dc	pF	
Min	75	100	0.85			2.0
Max		300	1.20	0.5	20	6.0

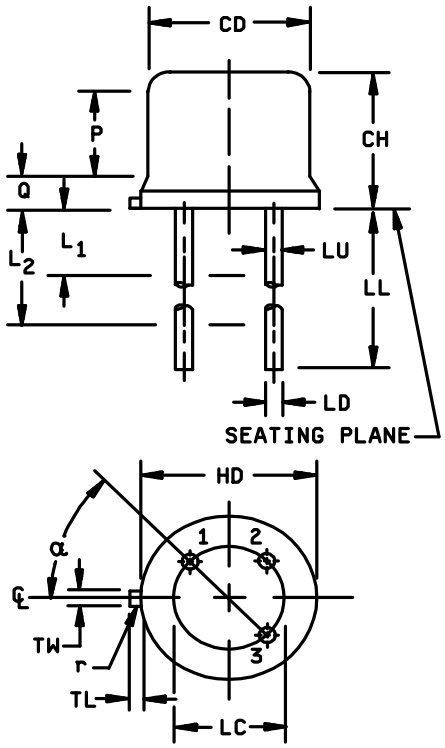
1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min.	Max.	Min.	Max.	
CD	.305	.335	7.75	8.51	
CH	0.24	0.26	6.1	6.6	
HD	.335	.37	8.51	9.4	
LC	.200 TP		5.08 TP		4
LD	.016	.021	.41	.53	2, 5
LL	.500	.750	12.70	19.05	5
LU	.016	.019	.41	.48	3, 5
L1		.050		1.27	
L2	.250		6.35		
P	.100		2.54		
Q		.040		1.02	
r		.007		.18	
TL	.029	.045	.74	1.14	
TW	.028	.034	.71	.86	
α	45° TP		45° TP		

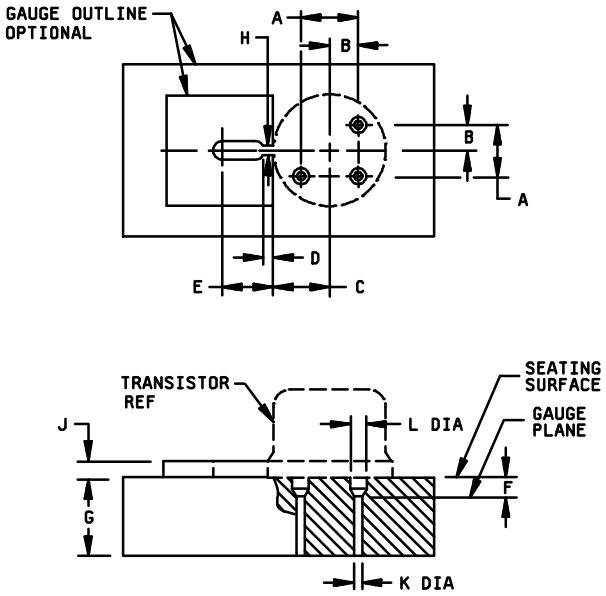


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 (6.35) from the seating plane.
4. Measured in the zone .050 (1.27 mm) and .250 (6.35) from the seating plane.
5. Measured from the maximum diameter of the actual case.
6. All three leads.
7. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
8. Lead 1 is the emitter, lead 2 is the base, and lead 3 is the collector which is electrically connected to the case

FIGURE 1. Physicval dimensions (similar to T0-39).

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.1409	.1419	3.579	3.604
B	.0702	.0712	1.783	1.809
C	.182	.199	4.62	5.05
D	.009	.011	0.23	0.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	0.889	0.902
J	.150 Nom		3.81 Nom	
K	.0325	.0335	0.826	0.851
L	.0595	.0605	1.511	1.537



NOTES:
1. Metric equivalents are given for general information only.
2. Dimensions are in inches.
3. The following gauging procedures shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.254 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.
4. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.

FIGURE 2. Gauge for lead and tab location.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to T0-39) herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked in accordance with MIL-PRF-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the transistor.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.3 Screening. Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX level only
9	Not applicable
11	h_{FE2} and I_{CBO1}
12	See 4.3.1
13	Subgroup 2 of table I herein, ΔI_{CBO1} = 100% of initial value or 10 nA dc, whichever is greater; Δh_{FE2} = ±20% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

2N4405 - - - - $V_{CB} \geq 30$ V dc; T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; P_T = 1.0 W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIb (JAN, JANTX), of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table VIb (JAN and JANTX) of MIL-PRF-19500.

Subgroup	Method	Condition
3	1027	$V_{CB} \geq 30$ V dc, P_T = 1.0 W, T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
2	2036	Test condition E.
6	1026	$V_{CB} \geq 30$ V dc, P_T = 1.0 W, T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 10 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 80 V dc	I_{CBO1}		10	µA dc
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	I_{CBO2}		25	nA dc
Emitter to base breakdown voltage	3061	Bias condition D, V_{EB} = 5.0 V dc	I_{EBO1}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 3.0 V dc	I_{EBO2}		25	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 0.1 mA dc; pulsed (see 4.5.1)	h_{FE1}	75		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE2}	100		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE3}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE4}	50		
Collector-emitter saturated voltage	3071	I_C = 10 mA dc; I_B = 1.0 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		0.15	V dc
Collector-emitter saturated voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		0.20	V dc
Collector-emitter saturated voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)3}$		0.50	V dc
Base-emitter non-saturated	3066	Test condition B; V_{CE} = 5 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	V_{BE}		0.90	V dc
Base-emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		0.80	V dc
Base-emitter saturated voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$	0.85	1.20	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	I_{CBO3}		25	µA dc
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 150 mA dc pulsed (see 4.5.1)	h_{FE5}	40		
Subgroup 4						
Delay time		V_{CC} = 31.9 V dc; $V_{BE(OFF)}$ = 0.8 V dc; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see figure 3)	t_d		15	ns
Rise time		V_{CC} = 31.9 V dc; $V_{BE(OFF)}$ = 0.8 V dc; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see figure 3)	t_r		25	ns
Storage time		V_{CC} = 31.9 V dc; I_C = 500 mA dc; I_{B1} = I_{B2} = 50 mA dc; (see figure 3)	t_s		175	ns
Fall time		V_{CC} = 31.9 V dc; I_C = 500 mA dc; I_{B1} = I_{B2} = 50 mA dc; (see figure 3)	t_f		50	ns
Magnitude of common emitter, small- signal short-circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 50 mA dc; f = 100 MHz	$\lvert h_{FE} \rvert$	2.0	6	
Noise figure	3246	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz; R_s 100 ohms	NF		3.5	dB
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0 mA dc 100 kHz ≤ f ≤ 1 MHz	C_{obo}		20	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; V_{CE} = 10 V dc; I_C = 0.5 A dc; t = 1 s; 1 cycle				
Electrical measurements		See table II, steps 1 and 3				

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		25	nA dc
2	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO4}		50	nA dc
3	Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE2}	100		
4	Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	Δh_{fe}	≯20 percent change		

1/ The electrical measurements for table VIb (JAN, JANTX) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 3.

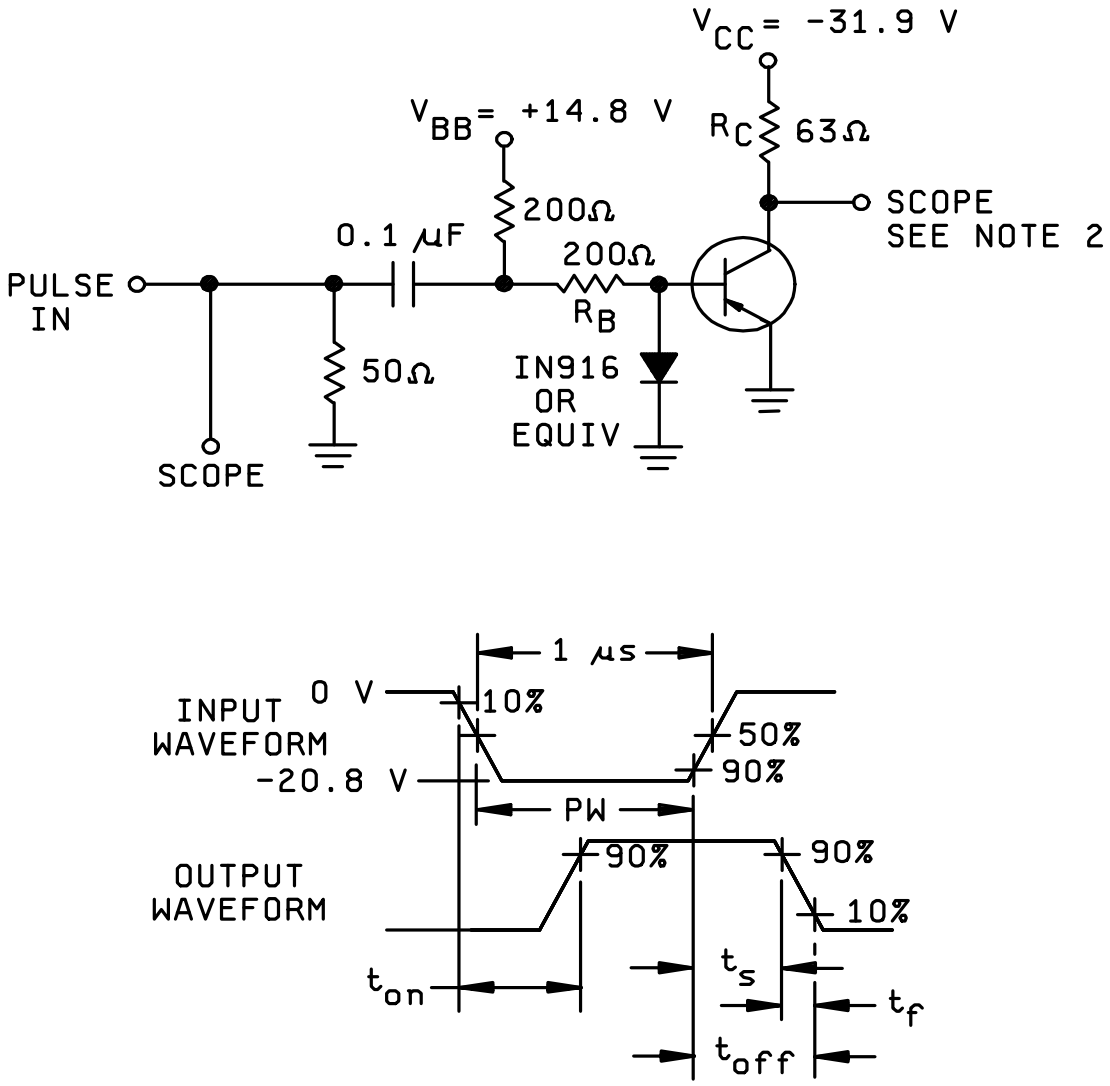
b. Subgroups 3 and 6, see table II herein, steps 2 and 4.

2/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1 and 3.

b. Subgroup 6, see table II herein, steps 2 and 4.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 ohms.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 0.2 ns.
3. To obtain data for curves, voltage levels are approximately as shown, R_B and R_C are varied.

FIGURE 3. Pulse response test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1.1 and 2.2).

b. The lead finish as specified (see 3.3.1).

c. For die acquisition, specify the JANHC or JANKC letter version (see figure 3).

d. Type designation and quality assurance level.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 85

User activities:
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

(Project 5961-F152)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/448B	2. DOCUMENT DATE 2 February 1999

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER TYPE 2N4405 JAN, JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete

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